Liang Yue

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My research interests include complex 3D generation, part decomposition and printable assembly, world models, 3D vision, and computer graphics.

Education

Nanjing University, School of Electronic Science and Engineering · Ph.D. Student
2026.09–Now
Advisors: Xiaoxiao Long and Xun Cao. Research: printable 3D generation and assembly design.

Tsinghua University, Department of Computer Science and Technology · M.S. in Computer Science 2022–2025.06
Advisor: Song-Hai Zhang. Research: digital 3D indoor scene construction algorithms.

Peking University, School of Information Science and Technology · B.S. in Computer Science; Double Major in Physics 2018–2022.06

Recent Research and Projects

Printable 3D Generation and Assembly Design for Complex Models

2026.10–Now
Investigating part decomposition, interlocking connector design, and 3D printing for complex 3D models. The workflow connects generation or reconstruction, connector design, simulation, and physical assembly. The focus is on assembly feasibility, ease of assembly, connection reliability, and stability.

CAI-WM: World-Action Modeling

2026.06–2026.09
Contributed evaluation management, experimental result recording, and training-data production. In the September 18, 2026 snapshot of WorldArena 2.0 Track 1, the team model scored 70.07 overall, only 2.26 points behind the leader. It placed 5th in Trajectory Accuracy, 6th in Physics Adherence, and 9th in Aesthetic Quality among 81 models.
Leaderboard snapshot

Privacy–Utility Trade-offs in Life-Logging Video (RayNeo Period)

2026.03–2026.04
Co-authored *Position: Life-Logging Video Streams Make the Privacy–Utility Trade-off Inevitable*, addressing privacy–utility trade-offs in persistent sensing and egocentric video systems. Second author, affiliated with RayNeo.AI in the manuscript; submitted to the NeurIPS 2026 Position Paper Track.
OpenReview

Internships and Research Experience

- **Horizon Robotics · Research Intern, Shanghai** 2026.05–2026.10. Contributed to world-action model development for autonomous driving.

- **RayNeo Smart Glasses (TCL) · Research Intern** 2025.10–2026.05. Worked on personal egocentric video assistants, memory construction, and retrieval. The privacy–utility paper above was completed during the RayNeo period.
- **Tsinghua University, College of Artificial Intelligence · Research Assistant** 2025.06–2026.03. Advisor: Miao Liu.
- **Kunlun Skyworks · Research Intern** 2025.07–2025.09. Fine-tuned a navigation world model using SEKAI and evaluated familiar, partially familiar, and unseen environments.

Publications

- **Synthesizing 3D Scenes via Diffusion Model that Incorporates Indoor Scene Characteristics.** Liang Yue, Shao-Kui Zhang, Lin Yuan, Yi-Tao Chen, Zi-Rui Zhou, Song-Hai Zhang. *ACM Multimedia*, 2025. DOI
- **SceneCluster: Interactive Scene Synthesis by Clustering Groups of Furniture Objects.** Shao-Kui Zhang, Liang Yue, Han-Xi Zhu, Hao-Ran Zhang, Song-Hai Zhang. *IEEE TVCG*, 2025. DOI
- **SceneExpander: Real-Time Scene Synthesis for Interactive Floor Plan Editing.** Shao-Kui Zhang, Junkai Huang, Liang Yue, Jia-Tong Zhang, Jia-Hong Liu, Yu-Kun Lai, Song-Hai Zhang. *ACM Multimedia*, 2024. DOI
- **A Survey on Digital 3D Indoor Scene Construction.** Liang Yue, Hao Tan, Junkai Huang, Shao-Kui Zhang. *Journal of Image and Graphics*, 2024.

Other Projects

- **Egocentric Video Fine-tuning for Multimodal LLMs** (from 2026.01): improved understanding and reasoning about first-person visual information.
- **Personal Egocentric Video Assistant** (from 2025.10): memory construction and retrieval for question answering.
- **Navigation World Model Fine-tuning** (2025.07–2025.09): training on new navigation data and evaluation across environments.
- **Field-based Indoor Scene Optimization** (2024.11–2025.04): combined physical constraints and layout plausibility into a field guiding object layout optimization.
- **Indoor Scene Semantics and Dataset Augmentation** (2024.07–2024.11): modeled object relationships and functional groups to expand 3D indoor scene datasets.

Skills and Research Interests

PyTorch, Transformers, Three.js, OpenGL, and Python / Flask. Research interests: 3D generation and printable assembly, world models, augmented reality, 3D vision, and computer graphics.